

Financial Mathematics

(MATH35400/MATHM5400)

Problem Class 6 – 27 February 2026 ¹

Problems:

1. (2022 exam, Q2 (b)) Consider the following model with two periods assuming that the interest rate is $r = 0$ and assume that u and d are arbitrary constants with $u > 6 > d$. We assume that at each point of time t an investor is aware of all prices up to that time point.

State	$t = 0$	$t = 1$	$t = 2$
ω_1	10	14	16
ω_2	10	14	12
ω_3	10	6	u
ω_4	10	6	d

An American *look-back put option* is an option where the holder is paid the amount $Y_t(\omega)$ where

$$Y_t(\omega) = (\max_{\nu: \nu \leq t} S_\nu(\omega) - S_t(\omega))^+.$$

- (a) **(4 marks)** Calculate the risk-neutral probability measure $\mathbb{Q}$ and specify $\mathbb{Q}(\{\omega_i\})$ for every $i = 1, 2, 3, 4$.
 - (b) **(2 marks)** Assume that $u = 11$ and $d = 5$ and determine the payoff Y_t of the American look-back put option for every state and every time point.
 - (c) **(5 marks)** Use risk-neutral valuation to find the time $t = 0$ value of the option assuming that $u = 11$ and $d = 5$.
 - (d) **(3 marks)** Decide whether an early exercise of the option at time point 1 is optimal in the case where $10 > u > 6 > d$.
2. (2021 exam, Q2a) Consider a model with three periods and one risky security where the interest rate is $r = 0.1$ and the price process is given by the following table:

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State	$t = 0$	$t = 1$	$t = 2$	$t = 3$
ω_1	10	16	20	24
ω_2	10	16	20	16
ω_3	10	16	18	20
ω_4	10	16	18	16
ω_5	10	8	12	16
ω_6	10	8	12	10
ω_7	10	8	6	8
ω_8	10	8	6	4

- (a) **(3 marks)** What does it mean for a random variable τ to be a stopping time with respect to a filtration $\{\mathcal{F}_t\}$?

State the definition of the Snell envelope Z_s of a stochastic process $\{X_t, t = 0, \dots, T\}$ adapted to some filtration $\{\mathcal{F}_t\}$.

State the optimal stopping rule at any time t , in terms of Z and X .

- (b) **(2 marks)** Determine the payoff process of an American put option with strike price $K = 10$ on one share of the stock.
- (c) **(7 marks)** Find the time 0 price for the American put option considered in ii. Determine an optimal stopping time and find a replicating portfolio for the trading strategy that uses this optimal stopping time.